

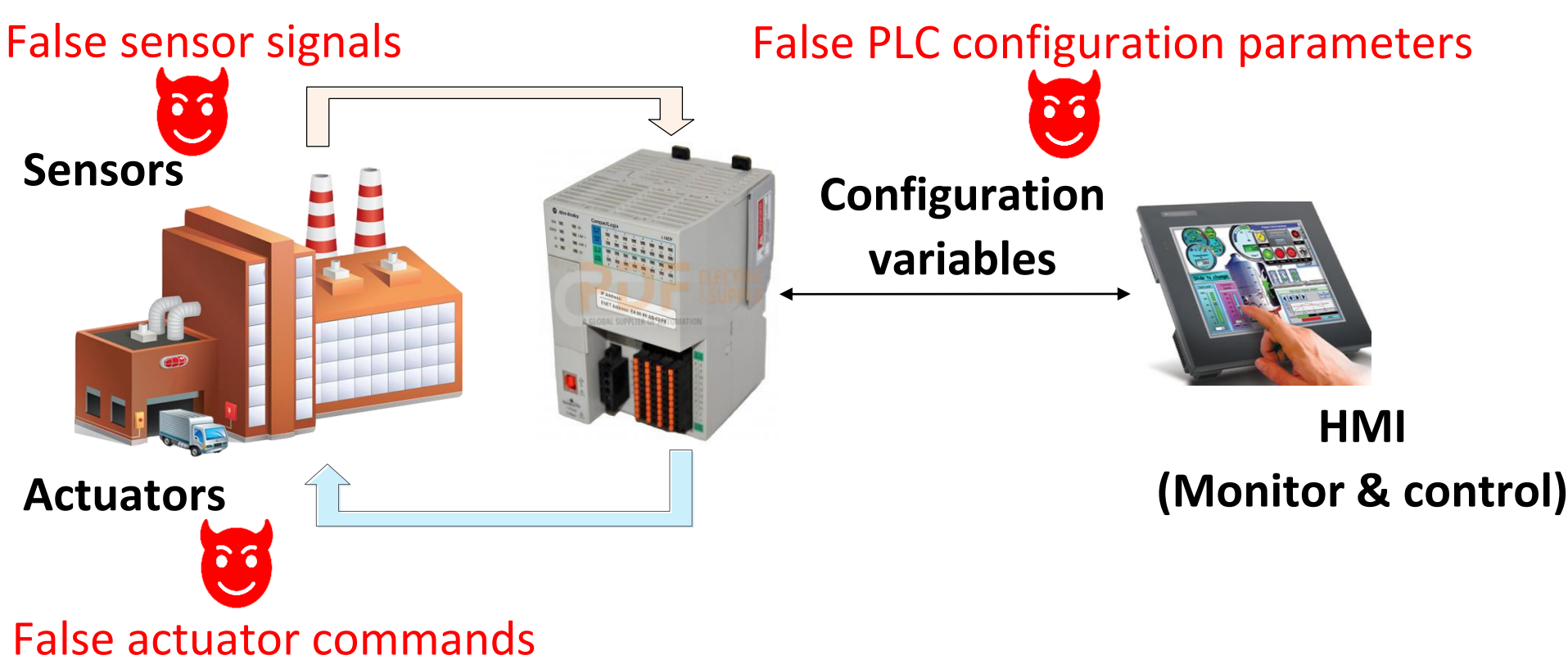
SAIN: Improving ICS Attack Detection Sensitivity via State-Aware Knowledge

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Introduction

In Industrial Control Systems (ICS), the PLC program takes sensor measurements as input and sends commands to actuators based on its configuration parameters.

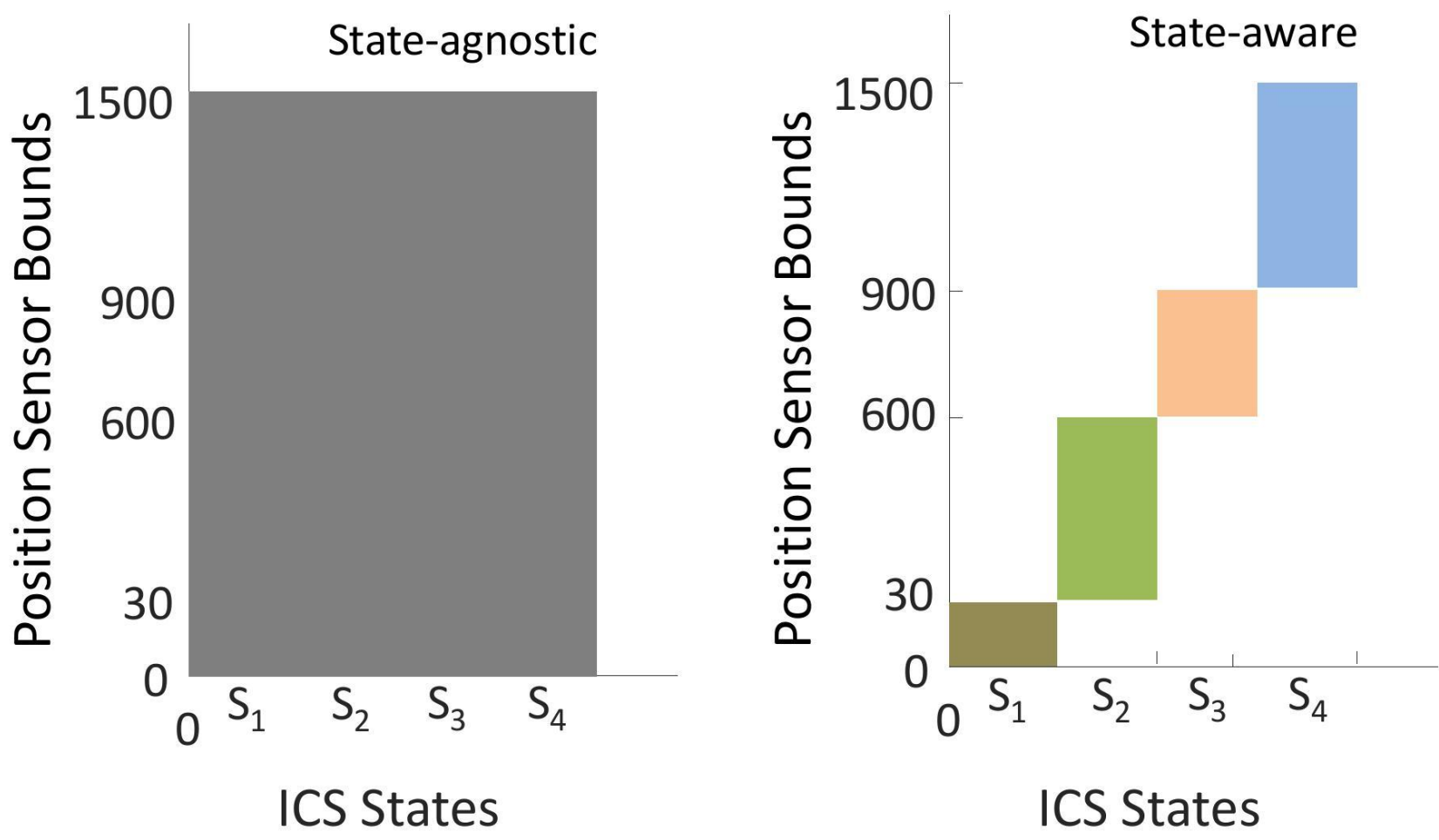
However, prior research has shown that ICSs are **vulnerable** to attacks that manipulate sensors, actuators, and configuration variables.



Motivation

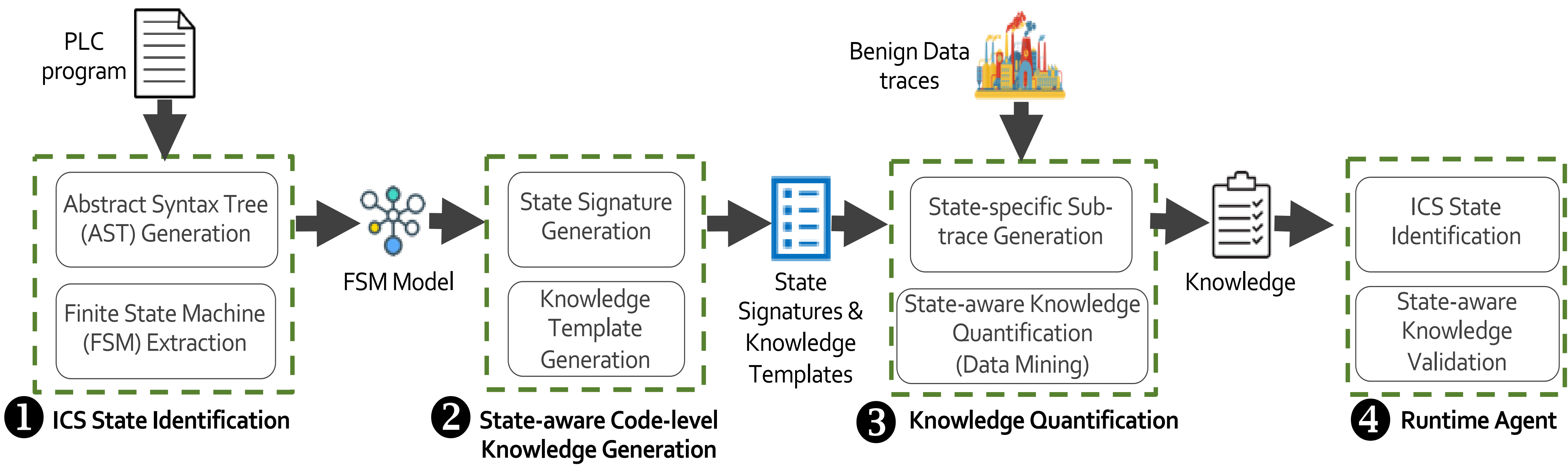
To detect such attacks, prior work has explored knowledge templates, which define the relationship between PLC variables and their value bounds.

Prior works are state-agnostic and define loose bounds that may allow stealthy attacks.

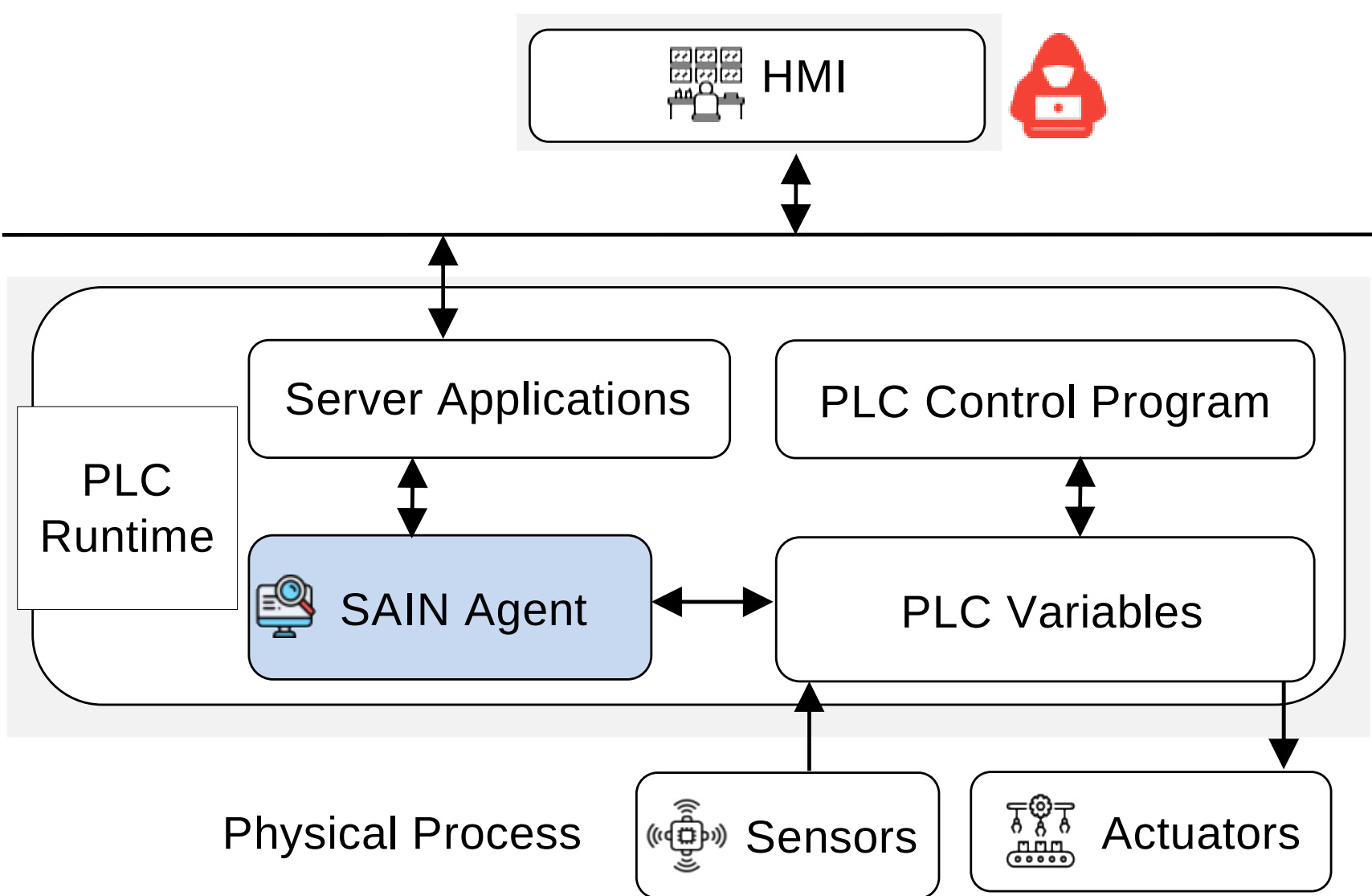


SAIN Overview

1. Identify the states implemented in the PLC program to control ICS operation.
2. Determine the conditions that must hold under each ICS state and convert them into knowledge templates.
3. Quantify the knowledge templates by leveraging ICS data traces partitioned by state.
4. Validate ICS knowledge upon receiving external requests through the HMI using a runtime agent.

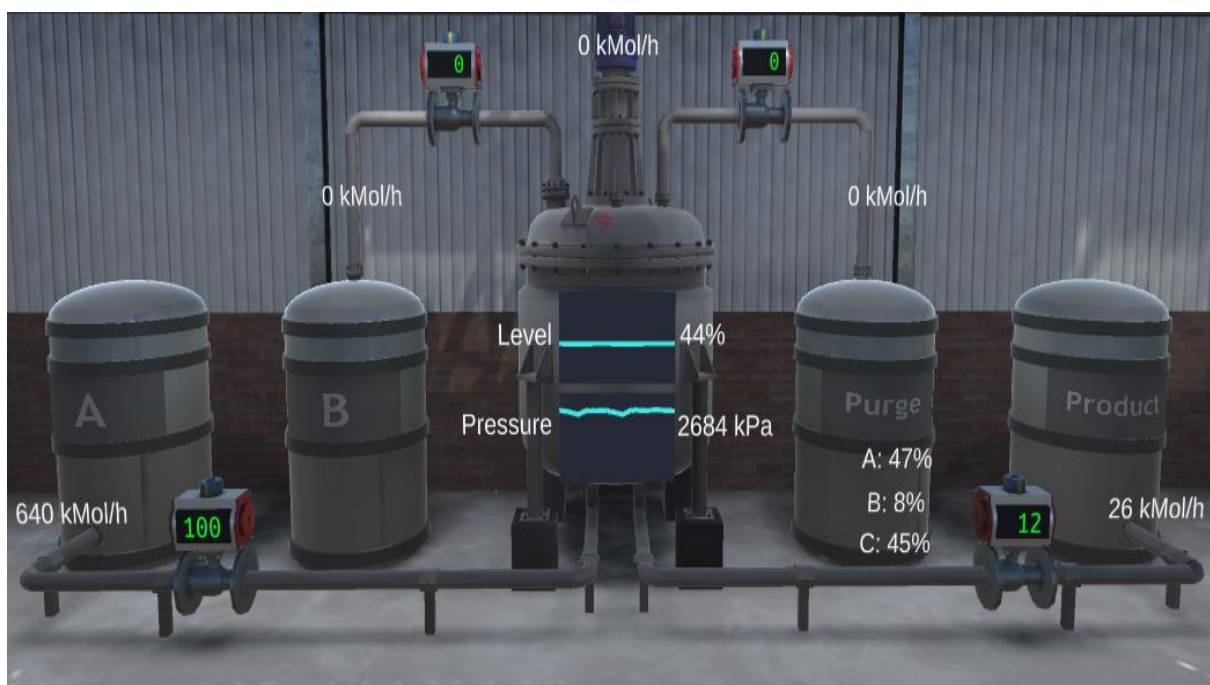
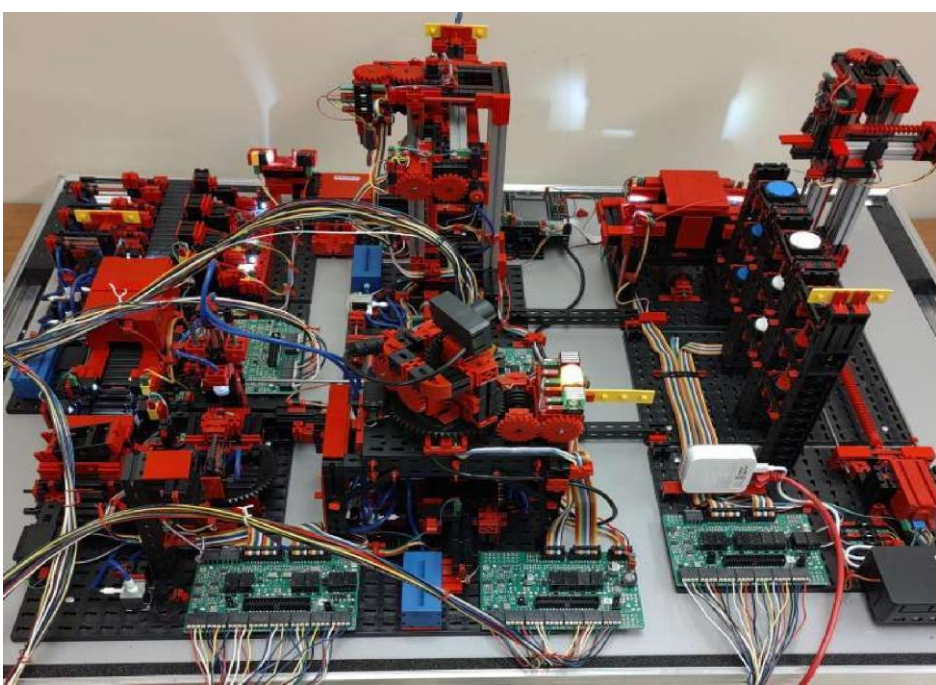


Runtime Detection Agent



Evaluation Results

We evaluated **SAIN** on two testbeds: a Fischertechnik manufacturing plant and a chemical plant.



ICS Attacks	Detection	Runtime Overhead
17	100%	3%